Analytic Federated Learning with SVD Compression

Implementation of Novelty.

Overview

What is AFL?

Analytic Federated Learning (AFL) is a single-round, gradient-free federated learning framework. Instead of iterative gradient descent, each client solves a closed-form least-squares problem using two local statistics — a feature covariance matrix C and a label correlation matrix W. These are aggregated on the server via the exact Woodbury matrix identity, producing a globally optimal solution in one communication round.

What is Novel?

Three novel contributions are evaluated across these experiments: (1) Automatic regularisation — a data-driven ridge fallback that prevents rank-deficient matrix inversion under extreme non-IID conditions; (2) Rank-r SVD compression of the local weight matrix W, reducing communication bandwidth at the cost of a small, controlled approximation error; and (3) Communication cost analysis, reported after every run to quantify savings in megabytes and percentage terms.

Experimental Setup:

All 16 result experiments use a pretrained ResNet-18 backbone, ridge coefficient rg = 1, regularisation cleaning enabled, and rank-r = 32 SVD compression. Heterogeneity is varied via the Dirichlet concentration parameter $\alpha \in \{0.1, 0.05\}$ and minimum shard sizes $S \in \{2, 4, 5, 10\}$. Scalability is tested with $K \in \{100, 500, 1000\}$ clients. Results are compared against the original paper's reported numbers shown in the comparison figure and table at the end of this document.

- **Section A — CIFAR-10 (Experiments 1–4)**

1. **CIFAR-10 with Alpha=0.1**

```
Aggregating!
Aggregation done!

==============================================================
  COMMUNICATION COST REPORT
==============================================================
  Embedding size (feat)   : 512
  Num classes             : 10
  Num clients (K)         : 100
  Rank-r compression      : 32 (effective = 10)
  C matrix dominance      : 98.1% of total per-client cost
--------------------------------------------------------------
                                     Original  Compressed
--------------------------------------------------------------
  W matrix  (per client)              0.0391MB     0.0399MB
  C matrix  (per client)              2.0000MB     2.0000MB
  Total     (per client)              2.0391MB     2.0399MB
--------------------------------------------------------------
  TOTAL     (all clients)           203.906MB    203.990MB
  W-only savings                       -2.1%
  Overall savings                      -0.0%  (-0.084 MB saved)

  NOTE: Low saving because C matrix (2.000 MB) dominates W.
        Use CIFAR-100/TinyImageNet for meaningful W compression gains.
        To show larger overall savings, compress C as well (future work).
==============================================================


Evaluating global model!
Total time (train + agg): 334.70s
Global Accuracy: 82.50%

Cleaning regularization...
Accuracy after cleaning: 82.49%
Total time (with cleaning): 352.42s
Saved results to cifar10_resnet18_100_0.1.csv
```

2. **CIFAR-10 with Alpha=0.05**

```
Aggregating!
Aggregation done!

==============================================================
  COMMUNICATION COST REPORT
==============================================================
  Embedding size (feat)   : 512
  Num classes             : 10
  Num clients (K)         : 100
  Rank-r compression      : 32 (effective = 10)
  C matrix dominance      : 98.1% of total per-client cost
--------------------------------------------------------------
                                     Original  Compressed
--------------------------------------------------------------
  W matrix  (per client)              0.0391MB     0.0399MB
  C matrix  (per client)              2.0000MB     2.0000MB
  Total     (per client)              2.0391MB     2.0399MB
--------------------------------------------------------------
  TOTAL     (all clients)           203.906MB    203.990MB
  W-only savings                       -2.1%
  Overall savings                      -0.0%  (-0.084 MB saved)

  NOTE: Low saving because C matrix (2.000 MB) dominates W.
        Use CIFAR-100/TinyImageNet for meaningful W compression gains.
        To show larger overall savings, compress C as well (future work).
==============================================================


Evaluating global model!
Total time (train + agg): 309.08s
Global Accuracy: 82.50%

Cleaning regularization...
Accuracy after cleaning: 82.49%
Total time (with cleaning): 325.68s
Saved results to cifar10_resnet18_100_0.05.csv
```

3. CIFAR-10 with Shard=4

```
Aggregating!
Aggregation done!

==============================================================
  COMMUNICATION COST REPORT
==============================================================
  Embedding size (feat)   : 512
  Num classes             : 10
  Num clients (K)         : 100
  Rank-r compression      : 32 (effective = 10)
  C matrix dominance      : 98.1% of total per-client cost
--------------------------------------------------------------
                                     Original  Compressed
--------------------------------------------------------------
  W matrix  (per client)              0.0391MB     0.0399MB
  C matrix  (per client)              2.0000MB     2.0000MB
  Total     (per client)              2.0391MB     2.0399MB
--------------------------------------------------------------
  TOTAL     (all clients)           203.906MB    203.990MB
  W-only savings                        -2.1%
  Overall savings                       -0.0%  (-0.084 MB saved)

  NOTE: Low saving because C matrix (2.000 MB) dominates W.
        Use CIFAR-100/TinyImageNet for meaningful W compression gains.
        To show larger overall savings, compress C as well (future work).
==============================================================


Evaluating global model!
Total time (train + agg): 308.58s
Global Accuracy: 82.50%

Cleaning regularization...
Accuracy after cleaning: 82.49%
Total time (with cleaning): 325.27s
Saved results to cifar10_resnet18_100_0.05.csv
```

4. CIFAR-10 with Shard=2

```
Aggregating!
Aggregation done!

==============================================================
  COMMUNICATION COST REPORT
==============================================================
  Embedding size (feat)   : 512
  Num classes             : 10
  Num clients (K)         : 100
  Rank-r compression      : 32 (effective = 10)
  C matrix dominance      : 98.1% of total per-client cost
--------------------------------------------------------------
                                     Original  Compressed
--------------------------------------------------------------
  W matrix  (per client)              0.0391MB     0.0399MB
  C matrix  (per client)              2.0000MB     2.0000MB
  Total     (per client)              2.0391MB     2.0399MB
--------------------------------------------------------------
  TOTAL     (all clients)           203.906MB    203.990MB
  W-only savings                        -2.1%
  Overall savings                       -0.0%  (-0.084 MB saved)

  NOTE: Low saving because C matrix (2.000 MB) dominates W.
        Use CIFAR-100/TinyImageNet for meaningful W compression gains.
        To show larger overall savings, compress C as well (future work).
==============================================================


Evaluating global model!
Total time (train + agg): 324.70s
Global Accuracy: 82.50%

Cleaning regularization...
Accuracy after cleaning: 82.49%
Total time (with cleaning): 340.97s
Saved results to cifar10_resnet18_100_0.1.csv
```

- **Section B — CIFAR-100 (Experiments 5–8)**

5. **CIFAR-100 with Alpha=0.1**

```
Aggregating!
Aggregation done!

==============================================================
  COMMUNICATION COST REPORT
==============================================================
  Embedding size (feat)   : 512
  Num classes             : 100
  Num clients (K)         : 100
  Rank-r compression      : 32 (effective = 32)
  C matrix dominance      : 83.7% of total per-client cost
--------------------------------------------------------------
                                     Original  Compressed
--------------------------------------------------------------
  W matrix  (per client)              0.3906MB     0.1497MB
  C matrix  (per client)              2.0000MB     2.0000MB
  Total     (per client)              2.3906MB     2.1497MB
--------------------------------------------------------------
  TOTAL     (all clients)            239.062MB    214.966MB
  W-only savings                        61.7%
  Overall savings                       10.1%  (24.097 MB saved)
==============================================================


Evaluating global model!
Total time (train + agg): 361.71s
Global Accuracy: 58.53%

Cleaning regularization...
Accuracy after cleaning: 58.60%
Total time (with cleaning): 380.10s
Saved results to cifar100_resnet18_100_0.1.csv
```

6. **CIFAR-100 with Alpha=0.05**

```
Local training time: 334.36s

Aggregating!
Aggregation done!

==============================================================
  COMMUNICATION COST REPORT
==============================================================
  Embedding size (feat)   : 512
  Num classes             : 100
  Num clients (K)         : 100
  Rank-r compression      : 32 (effective = 32)
  C matrix dominance      : 83.7% of total per-client cost
--------------------------------------------------------------
                                     Original  Compressed
--------------------------------------------------------------
  W matrix  (per client)              0.3906MB     0.1497MB
  C matrix  (per client)              2.0000MB     2.0000MB
  Total     (per client)              2.3906MB     2.1497MB
--------------------------------------------------------------
  TOTAL     (all clients)            239.062MB    214.966MB
  W-only savings                        61.7%
  Overall savings                       10.1%  (24.097 MB saved)
==============================================================


Evaluating global model!
Total time (train + agg): 354.51s
Global Accuracy: 58.64%

Cleaning regularization...
Accuracy after cleaning: 58.55%
Total time (with cleaning): 372.07s
Saved results to cifar100_resnet18_100_0.05.csv
```

7. CIFAR-100 Shard = 10

```
Local training time: 342.26s

Aggregating!
Aggregation done!

==============================================================
  COMMUNICATION COST REPORT
==============================================================
  Embedding size (feat)   : 512
  Num classes             : 100
  Num clients (K)         : 100
  Rank-r compression      : 32 (effective = 32)
  C matrix dominance      : 83.7% of total per-client cost
--------------------------------------------------------------
                                     Original  Compressed
--------------------------------------------------------------
  W matrix  (per client)              0.3906MB     0.1497MB
  C matrix  (per client)              2.0000MB     2.0000MB
  Total     (per client)              2.3906MB     2.1497MB
--------------------------------------------------------------
  TOTAL     (all clients)            239.062MB    214.966MB
  W-only savings                        61.7%
  Overall savings                       10.1%  (24.097 MB saved)
==============================================================


Evaluating global model!
Total time (train + agg): 362.51s
Global Accuracy: 58.43%

Cleaning regularization...
Accuracy after cleaning: 58.45%
Total time (with cleaning): 380.24s
Saved results to cifar100_resnet18_100_0.1.csv
```

8. CIFAR-100 Shard = 5

```
Local training time: 340.86s

Aggregating!
Aggregation done!

==============================================================
  COMMUNICATION COST REPORT
==============================================================
  Embedding size (feat)   : 512
  Num classes             : 100
  Num clients (K)         : 100
  Rank-r compression      : 32 (effective = 32)
  C matrix dominance      : 83.7% of total per-client cost
--------------------------------------------------------------
                                     Original  Compressed
--------------------------------------------------------------
  W matrix  (per client)              0.3906MB     0.1497MB
  C matrix  (per client)              2.0000MB     2.0000MB
  Total     (per client)              2.3906MB     2.1497MB
--------------------------------------------------------------
  TOTAL     (all clients)            239.062MB    214.966MB
  W-only savings                        61.7%
  Overall savings                       10.1%  (24.097 MB saved)
==============================================================


Evaluating global model!
Total time (train + agg): 361.75s
Global Accuracy: 58.43%

Cleaning regularization...
Accuracy after cleaning: 58.45%
Total time (with cleaning): 378.43s
Saved results to cifar100_resnet18_100_0.1.csv
```

- **Section C — Tiny-ImageNet (Experiments 9–12)**

9. Tinyimagenet with Alpha=0.1

```
Local training time: 633.21s

Aggregating!
Aggregation done!

==============================================================
  COMMUNICATION COST REPORT
==============================================================
  Embedding size (feat)   : 512
  Num classes             : 200
  Num clients (K)         : 100
  Rank-r compression      : 32 (effective = 32)
  C matrix dominance      : 71.9% of total per-client cost
--------------------------------------------------------------
                                     Original  Compressed
--------------------------------------------------------------
  W matrix  (per client)              0.7812MB     0.1741MB
  C matrix  (per client)              2.0000MB     2.0000MB
  Total     (per client)              2.7812MB     2.1741MB
--------------------------------------------------------------
  TOTAL     (all clients)           278.125MB    217.407MB
  W-only savings                        77.7%
  Overall savings                       21.8%  (60.718 MB saved)
==============================================================


Evaluating global model!
Total time (train + agg): 655.86s
Global Accuracy: 54.98%

Cleaning regularization...
Accuracy after cleaning: 55.06%
Total time (with cleaning): 673.95s
Saved results to tinyimagenet_resnet18_100_0.1.csv
```

10. Tinyimagenet with Alpha=0.05

```
Local training time: 619.41s

Aggregating!
Aggregation done!

==============================================================
  COMMUNICATION COST REPORT
==============================================================
  Embedding size (feat)   : 512
  Num classes             : 200
  Num clients (K)         : 100
  Rank-r compression      : 32 (effective = 32)
  C matrix dominance      : 71.9% of total per-client cost
--------------------------------------------------------------
                                     Original  Compressed
--------------------------------------------------------------
  W matrix  (per client)              0.7812MB     0.1741MB
  C matrix  (per client)              2.0000MB     2.0000MB
  Total     (per client)              2.7812MB     2.1741MB
--------------------------------------------------------------
  TOTAL     (all clients)           278.125MB    217.407MB
  W-only savings                        77.7%
  Overall savings                       21.8%  (60.718 MB saved)
==============================================================


Evaluating global model!
Total time (train + agg): 642.41s
Global Accuracy: 54.74%

Cleaning regularization...
Accuracy after cleaning: 54.80%
Total time (with cleaning): 661.06s
Saved results to tinyimagenet_resnet18_100_0.05.csv
```

11. Tiny-ImageNet with Shard = 10

```
Local training time: 623.93s

Aggregating!
Aggregation done!

==============================================================
  COMMUNICATION COST REPORT
==============================================================
  Embedding size (feat)   : 512
  Num classes             : 200
  Num clients (K)         : 100
  Rank-r compression      : 32 (effective = 32)
  C matrix dominance      : 71.9% of total per-client cost
--------------------------------------------------------------
                                     Original  Compressed
--------------------------------------------------------------
  W matrix  (per client)              0.7812MB     0.1741MB
  C matrix  (per client)              2.0000MB     2.0000MB
  Total     (per client)              2.7812MB     2.1741MB
--------------------------------------------------------------
  TOTAL     (all clients)            278.125MB    217.407MB
  W-only savings                        77.7%
  Overall savings                       21.8%  (60.718 MB saved)
==============================================================


Evaluating global model!
Total time (train + agg): 647.33s
Global Accuracy: 54.97%

Cleaning regularization...
Accuracy after cleaning: 54.97%
Total time (with cleaning): 669.09s
Saved results to tinyimagenet_resnet18_100_0.1.csv
```

12. Tiny-ImageNet with Shard = 5

```
Local training time: 631.81s

Aggregating!
Aggregation done!

==============================================================
  COMMUNICATION COST REPORT
==============================================================
  Embedding size (feat)   : 512
  Num classes             : 200
  Num clients (K)         : 100
  Rank-r compression      : 32 (effective = 32)
  C matrix dominance      : 71.9% of total per-client cost
--------------------------------------------------------------
                                     Original  Compressed
--------------------------------------------------------------
  W matrix  (per client)              0.7812MB     0.1741MB
  C matrix  (per client)              2.0000MB     2.0000MB
  Total     (per client)              2.7812MB     2.1741MB
--------------------------------------------------------------
  TOTAL     (all clients)            278.125MB    217.407MB
  W-only savings                        77.7%
  Overall savings                       21.8%  (60.718 MB saved)
==============================================================


Evaluating global model!
Total time (train + agg): 656.15s
Global Accuracy: 54.97%

Cleaning regularization...
Accuracy after cleaning: 54.97%
Total time (with cleaning): 676.91s
Saved results to tinyimagenet_resnet18_100_0.1.csv
```

- **Section D — Scalability K=500/1000 (Experiments 13–16)**

13. CIFAR-100 with No.of Cleints (K) = 500

```
Local training time: 536.41s

Aggregating!
Aggregation done!

============================================================
  COMMUNICATION COST REPORT
============================================================
  Embedding size (feat)   : 512
  Num classes             : 100
  Num clients (K)         : 500
  Rank-r compression      : 32 (effective = 32)
  C matrix dominance      : 83.7% of total per-client cost
------------------------------------------------------------
                                    Original  Compressed
------------------------------------------------------------
  W matrix  (per client)             0.3906MB     0.1497MB
  C matrix  (per client)             2.0000MB     2.0000MB
  Total     (per client)             2.3906MB     2.1497MB
------------------------------------------------------------
  TOTAL     (all clients)          1195.312MB   1074.829MB
  W-only savings                       61.7%
  Overall savings                      10.1%  (120.483 MB saved)
============================================================


Evaluating global model!
Total time (train + agg): 571.25s
Global Accuracy: 58.37%

Cleaning regularization...
Accuracy after cleaning: 58.50%
Total time (with cleaning): 591.70s
Saved results to cifar100_resnet18_500_0.1.csv
```

14. CIFAR-100 with No.of Cleints (K) = 1000

```
Aggregating!
Aggregation done!

============================================================
  COMMUNICATION COST REPORT
============================================================
  Embedding size (feat)   : 512
  Num classes             : 100
  Num clients (K)         : 1000
  Rank-r compression      : 32 (effective = 32)
  C matrix dominance      : 83.7% of total per-client cost
------------------------------------------------------------
                                    Original  Compressed
------------------------------------------------------------
  W matrix  (per client)             0.3906MB     0.1497MB
  C matrix  (per client)             2.0000MB     2.0000MB
  Total     (per client)             2.3906MB     2.1497MB
------------------------------------------------------------
  TOTAL     (all clients)          2390.625MB   2149.658MB
  W-only savings                       61.7%
  Overall savings                      10.1%  (240.967 MB saved)
============================================================


Evaluating global model!
Total time (train + agg): 798.11s
Global Accuracy: 58.21%

Cleaning regularization...
Accuracy after cleaning: 58.62%
Total time (with cleaning): 814.27s
Saved results to cifar100_resnet18_1000_0.1.csv
```

15. Tinyimagenet with No.of Cleints (K) = 500

```
Local training time: 928.11s

Aggregating!
Aggregation done!

==============================================================
  COMMUNICATION COST REPORT
==============================================================
  Embedding size (feat)   : 512
  Num classes             : 200
  Num clients (K)         : 500
  Rank-r compression      : 32 (effective = 32)
  C matrix dominance      : 71.9% of total per-client cost
--------------------------------------------------------------
                                     Original  Compressed
--------------------------------------------------------------
  W matrix  (per client)              0.7812MB     0.1741MB
  C matrix  (per client)              2.0000MB     2.0000MB
  Total     (per client)              2.7812MB     2.1741MB
--------------------------------------------------------------
  TOTAL     (all clients)           1390.625MB   1087.036MB
  W-only savings                        77.7%
  Overall savings                       21.8%  (303.589 MB saved)
==============================================================


Evaluating global model!
Total time (train + agg): 966.48s
Global Accuracy: 54.84%

Cleaning regularization...
Accuracy after cleaning: 54.94%
Total time (with cleaning): 988.81s
Saved results to tinyimagenet_resnet18_500_0.1.csv
```

16. Tinyimagenet with No.of Cleints (K) = 1000

```
Aggregating!
Aggregation done!

==============================================================
  COMMUNICATION COST REPORT
==============================================================
  Embedding size (feat)   : 512
  Num classes             : 200
  Num clients (K)         : 1000
  Rank-r compression      : 32 (effective = 32)
  C matrix dominance      : 71.9% of total per-client cost
--------------------------------------------------------------
                                     Original  Compressed
--------------------------------------------------------------
  W matrix  (per client)              0.7812MB     0.1741MB
  C matrix  (per client)              2.0000MB     2.0000MB
  Total     (per client)              2.7812MB     2.1741MB
--------------------------------------------------------------
  TOTAL     (all clients)           2781.250MB   2174.072MB
  W-only savings                        77.7%
  Overall savings                       21.8%  (607.178 MB saved)
==============================================================


Evaluating global model!
Total time (train + agg): 1269.52s
Global Accuracy: 54.94%

Cleaning regularization...
Accuracy after cleaning: 55.00%
Total time (with cleaning): 1288.84s
Saved results to tinyimagenet_resnet18_1000_0.1.csv
```